MILITARY SPECIFICATION

TURRET SUBSYSTEM, UNIVERSAL (UTS) FOR ARMAMENT SUBSYSTEM, HELICOPTER, 20MM: M97E1 AND M97E2

The specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains quality assurance provisions and preparation for delivery for Universal Turret Subsystem (UTS) for the Armament Subsystem, Helicopter, 20MM: M97E1 and M97E2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	Electronic Equipment, Airborne, General Specification for.
MIL-E-6051	Electromagnetic Compatibility Requirements, Systems.
MIL-W-63150	Weapons and Support Materiel, Standard Quality Assurance Provisions for.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking For Shipment and Storage.
MIL-STD-410	-	Nondestructive Testing Personnel Qualification and Certification.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-704	-	Aircraft Electric Power Characteristics.
MIL-STD-810	-	Environmental Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

11830963	-	Turret Subsystem, Universal

PACKAGING DATA SHEET

P11830963	-	Turret Subsystem, Universal

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

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3.3 Universal turret subsystem (UTS). The UTS shall comply with all requirements specified on Drawing (Dwg). F11830963, all associated drawings, and with all requirements specified in applicable specifications and standards. The UTS covered by this specification shall be configured for use on AH-IS series helicopters. The UTS is an electric-powered turret which shall accommodate the M197 (20MM) gun. A simplified electrical block diagram is shown in Figure 1.

3.3.1 Major component list. Major components of the UTS are listed in Table I.

TABLE I. Major components.

TITLE	DRAWINGS
Turret Assembly	F11830962
Gun Drive Assembly	F198F734
Recoil Adapter Assembly	D11830567
Gun Control Assembly	F11830418
Logic Control Assembly	F11830922
Turret Control Assembly	F11830908
Storage/Feed Assembly - 20MM	F11830886
Drive Motor Assembly	D905D490
Pressure Transducer Assembly	C116C3555
Slider Assembly	C11830565

3.4 Characteristics.

3.4.1 Performance.

3.4.1.1 Positioning accuracy. The turret shall assume a position not to exceed 3 milliradians (mr) in the azimuth and elevation axes from a command static position for any position within the turret angular coverage.

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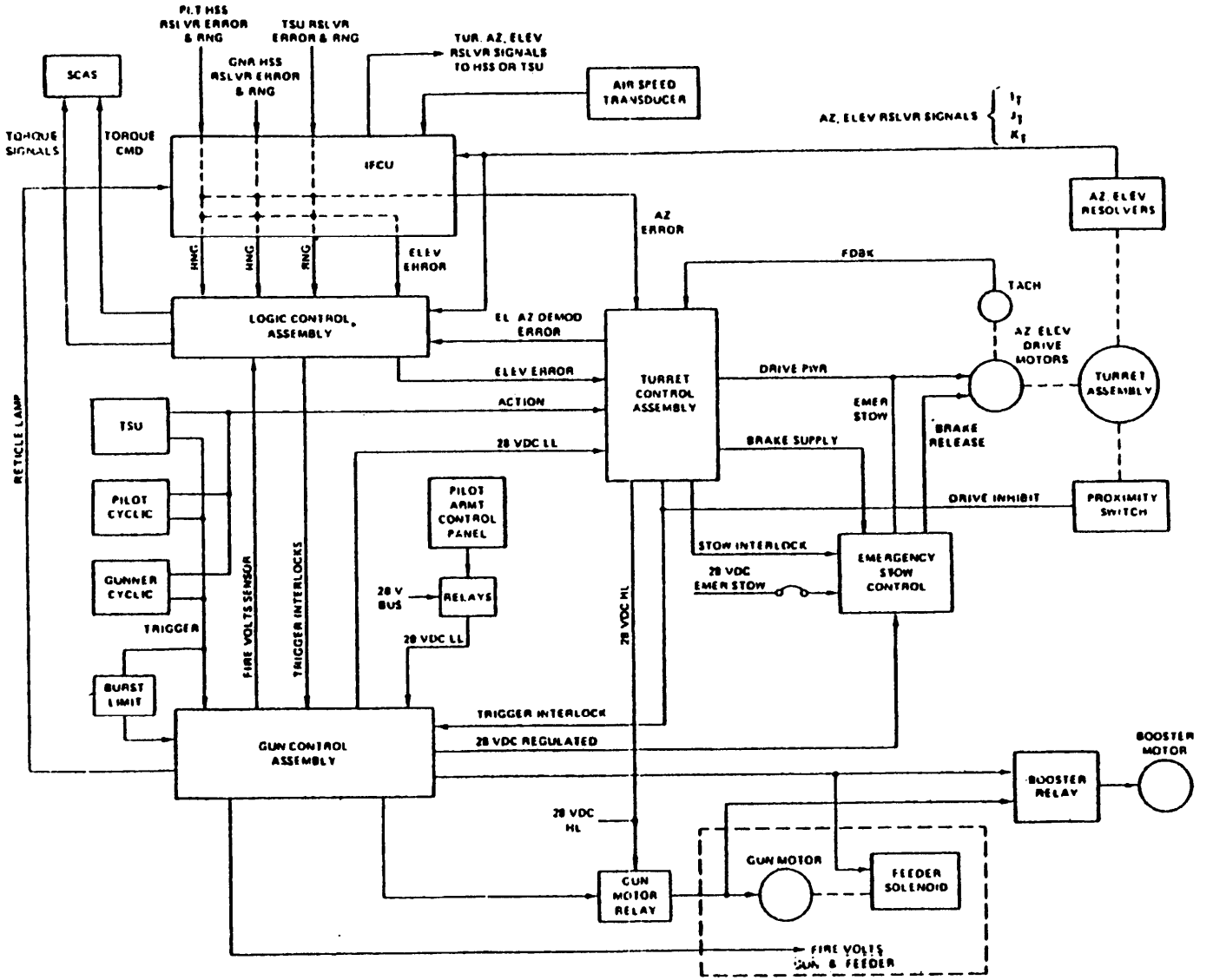


Figure 1. Turret Subsystem Simplified Block Diagram

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3.4.1.3 Firing rates. The turret subsystem shall have a firing rate of 730 + 50 shots per minute (SPM) under standard conditions.

3.4.1.4 Turret angular coverage.

3.4.1.4.1 Azimuth. The turret shall traverse electrically 110 ± 0.5 degrees to the right and 110 ± 0.5 degrees to the left of the helicopter butt line 0.00 straight forward position. The turret rotation shall be limited by electrical stops set for a maximum of 3 degrees before reaching the mechanical stops. The mechanical stops shall prevent turret rotation beyond 113 degrees to the right and left at full turret traverse velocity.

3.4.1.4.2 Elevation/deparession. The turret shall traverse the weapon to the maximum elevation that can be allowed in reference to the airframe and the rotor tip path plane (see Table II for devation limits versus azimuth position). Electrical limits shall be provided in accordance with Table II. Mechanical stops shall prevent turret movement beyond the maximum elevation limits of Table II at full turret traverse velocity. Both elevation and depression shall be measured from a plane parallel with the turret mounting surface. Mechanical and electrical stops shall be provided for depression. Depression angular travel for the weapon systems shall be 50 degrees minimum.

TABLE II. Turret elevation limits.

Degrees Azimuth	Degrees Elevation	
	Mechanical	Electrical
0°	17.25°	14.75°
10°	18.43°	15.93°
20°	20.50°	18.00°
30°	21.40°	18.90°
40°	23.00°	20.50°
56°	23.00°	20.50°
Straight line to 110°	16.50°	13.10°

NOTE: Mechanical limits shown are maximum. Electrical limit tolerance is ± 0.5 degree and must always activate prior to contacting mechanical limit.

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3.4.1.5 Turret velocity and acceleration.

3.4.1.5.1 Azimuth slew rate. The turret azimuth slew rate shall be a minimum of 60 degrees per second. Full azimuth rotation of 220 degrees left or right shall be a maximum of 4.2 seconds. Operation of the turret at any slew rate shall be smooth and not result in any oscillatory or erratic movements.

3.4.1.5.2 Elevation/depression slew rate. The turret elevation and depression slew rate shall be a minimum of 50 degrees per second. The maximum travel (elevation to depression or depression to elevation) of at least 65 degrees shall be attained in a maximum of 1.8 seconds. Operation of the turret at any slew rate shall be smooth and not result in any oscillatory or erratic movement.

3.4.1.5.3 Azimuth velocity error. With the azimuth command signal to the turret changing at a constant rate corresponding to a gun movement of 5 degrees per second in azimuth, the position error shall not exceed 2 milliradians.

3.4.1.5.4 Elevation velocity error. With the elevation command signal to the turret changing at a constant rate corresponding to gun movement of 5 degrees per second in elevation, the position error shall not exceed 2 milliradians.

3.4.1.5.5 Slew acceleration. The UTS shall accelerate the turret at a minimum rate of 120 degrees/sec^2 in any axis or combination of axes. The acceleration shall be determined using a step input to accelerate the turret from a standstill condition. Operation of the turret at maximum slew rate shall not result in any oscillatory or erratic movements.

3.4.1.5.6 Functional check-slew rate. During functional check or boresighting, at the operator's discretion, a signal with reference to ground shall be supplied to the turret. As long as this signal is present, the turret/gun azimuth and elevation slew rates shall be limited to a 6 $\pm$ 2 degrees per second velocity. This signal and slew rate shall have no adverse effect on boresight accuracy.

3.4.1.6 Weapon stowage. The UTS shall provide the capability for:

a. Fixed forward firing

b. Stow the gun during flight and during landing for ground clearance

c. Stow the gun in the event of loss of system power at speeds up to 100 knots.

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3.4.1.6.1 Fixed forward firing. A fixed forward firing position shall be provided in which the gun will be positioned at 0 ± 0.5 degree azimuth with respect to helicopter buttock line 0.00 and 0 ± 0.5 degree elevation relative to the turret mounting surface in accordance With Drawing 11830963. The turret shall assume the fixed forward firing position upon command from the pilot or gunner and shall be capable of being boresighted in this position.

3.4.1.6.2 Normal stow. The normal stow position for the UTS shall be 0 ± 1 degree in azimuth and 11 ± 1 degree in elevation relative to turret mounting surface. Whenever the UTS action switches are released or there is no fixed forward firing command, the weapon shall assume and maintain the normal stow position.

3.4.1.6.3 Emergency stow. In the event of an aircraft or UTS power failure, the turret shall automatically return to 0 ± 5 degrees in azimuth and a minimum of +10 degrees in elevation from the straight forward position at speeds up to 100 knots. The position shall be maintained under all flight conditions. The UTS shall provide a 28 Vdc signal to a panel indicator when the turret is in the emergency stow condition.

3.4.1.6.4 Stow locks. The stow locks or mechanisms shall hold the turret or gun in the stow position. They shall not hinder the movement of the UTS whenever the power drive is in operation. A method for manually releasing these locks shall be provided for ground servicing.

3.4.1.7 Firing interruption. The UTS gun firing circuit shall be interrupted and the weapon cleared without firing when any of the following conditions exist:

a. After any actuation command to fire any fixed forward firing wing store.

b. The boreline of the firing barrel of the weapon differs from the commanded boreline position in either azimuth or elevation by more than 1.5 degrees, including 180 degrees out-of-concidence.

c. Whenever the weapon is in stow or commanded to the normal stow position as described in 3.4.1.6.2, the weapon firing circuits shall be disabled.

3.4.1.7.1 Wing stores safety interlock. A safety interlock shall be included in the UTS to prevent wing stores firing unless the gun is Within 5 degrees of the O degree azimuth position.

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3.4.1.7.2 Firing interrupion indication. Whenever the gun firing circuit is interrupted as described in 3.4.1.7, the UTS shall provide a pulsed signal with reference to ground to the sighting system, 180 pulses per minute at a 50 percent duty cycle. When the gun is not in an interrupt mode, this signal shall be a steady ground signal.

3.4.1.7.3 Time delay. A time delay of 1 ± 0.5 seconds shall be incorporated in the UTS firing interruption circuit before gun firing can be resumed after release of the RMS trigger switch.

3.4.1.8 Rounds counter. The turret subsystem shall provide a signal pulse for each round cycled through it. This signal shall be-used-to actuate a helicopter cockpit indicator (which shows the number of rounds remaining in the ammunition feed system) and the turret mounted rounds totalizer of 3.4.1.8.1 (which shows total number of rounds cycled through the turret assembly). The counting signal accuracy shall be correct to within + 3 percent of actual number of rounds cycled. (The cockpit indicator has a requirement of being correct to within 50 rounds per complement; however, the actual amount of ammunition remaining shall not be less than the amount indicated.) The sensor that generates the signal pulse shall sense all UTS usage including live rounds, dummy rounds, and dry cycling.

3.4.1.8.1 Rounds totalizer. The rounds totalizer shall indicate the total number of rounds cycled through the subsystem. The rounds totalizer shall act independently of the gun. This rounds totalizer shall be non-resetable.

3.4.2 Weight. The components of the UTS, excluding gun and ammunition, shall not exceed the weight limits specified in below.

Title	Drawing	Weight Limit (pounds)
Turret Assembly	11830962	179.0
Electronic Components		28.8
Gun Control	11830418	
Logic Control	11830922	
Turret Control	11830908	
Pressure Transducer	116C3555	0.8

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Feed System	11830866 With Spring MS24586-815	93.9
Mechanical Components		23.8
Recoil Adapter	11830567	
Slider	11830565	
Gun Drive Assembly	189F134	
Bolts (8 used)	MS21250-06004	

3.4.3 Boresight. Boresighting shall be conducted using the provisions for azimuth and elevation on the UTS and it shall be performed by 2 persons in 15 minutes or less excluding aircraft set-up time.

3.4.4 Reliability (first article inspection only). The UTS, excluding Government furnished guns, ammunition links and other hardware; shall have a minimum acceptable mean-missions-between-failures of not less than 10 missions. A reliability mission is the expenditure of 750 rounds of 20MM ammo during a 1.4 hour flight. These requirements shall be demonstrated to a 95 percent confidence. In order to demonstrate this requirement, a minimum of 23,000 rounds will have to be fired.

3.4.5 Environmental conditions (first article testing only). The UTS shall operate and meet the performance requirements of 3.4.1 under any of the environmental conditions, or reasonable combination of these conditions, as specified below:

3.4.5.1 Natural environment. UTS components shall operate and withstand storage in the worldwide conditions as listed below:

3.4.5.1.1 Altitude. Operating 0-20,000 feet; nonoperating and storage 0-50,000 feet.

3.4.5.1.2 Temperature. Operating -65° to +125°F, non-operating, and storage -65° to +160°F.

3.4.5.1.3 Humidity. Operating in relative humidity up to 100 percent including conditions where condensation is in the form of water and frost.

3.4.5.1.4 Salt atmosphere. Operation after being subjected to MIL-STD-810, Method 509.1 for 48 hours.

3.4.5.1.5 Sand and dust. Operation after being subjected to MIL-STD-810, Method 510.1 for 6 hours.

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3.4.5.1.6 Fungus. Operating, nonoperating, and storage after being subjected to fungus specified below:

FUNGUS GROUP	SERIAL NUMBER
Aspergillus niger	9642
Aspergillus flavus	9643
Aspergillus versieolor	11730
Penecillium funiculosom	11797
Chactomium globosum	6205

3.4.5.1.7 Explosive atmosphere. Operation in explosive vapor within or surrounding the equipment.

3.4.5.1.8 Rain. The external-mounted components, including the gun control unit, shall operate without degradation in specified performance and shall sustain no physical damage during periods of heavy precipitation.

3.4.5.2 Service environment. The UTS electronic control units shall withstand the service environment as specified in MIL-E-5400 for Class IA equipment.

3.4.5.3 Induced environment.

3.4.5.3.1 - Vibration.

3.4.5.3.1.1 Electronic control units. The electronic control units shall not sustain damage or fail to provide specified performance when subjected to the vibration environment defined by MIL-E-5400, Figure 5 (sheet 1), Curve III.

3.4.5.3.1.2 Gun turret. The gun turret, with all turret-mounted electrical/electronic components and with any of the weapons installed, shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration acceleration levels specified in 4.5.2.1.

3.4.5.3.2 shock;

3.4.5.3.2.1 Gun turret and electronic control units. The gun turret (with all turret-mounted electrical/electronic components and the-weapon installed), and the electronic control units shall not sustain damage or subsequently fail to provide specified performance when subjected to 18 impact shocks with an amplitude of 15g, consisting of 3 shocks in opposite directions along each of 3 mutually-perpendicular axes, each shock impulse having a time duration of 11 $\pm$ 1 milliseconds.

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3.4.5.3.2.2 Bench handling. The electronic control units shall withstand shock environments encountered in maintenance and servicing.

3.4.6 Transportability. The UTS shall withstand vibration environments encountered during transporting.

3.4.7 Electromagnetic compatibility (EMC). Each UTS component, shall not cause undesirable response, malfunction, or degraded performance to any other equipment/subsystem installed or associated with the aircraft; nor, shall any component malfunction, exhibit degraded performance or undesirable response beyond the tolerances specified in MIL-E-6051 when other equipment on board the aircraft is singularly or collectively operated and performing their intended functions. Components tested in laboratory configuration shall meet the following requirements of MIL-STD-461: CEO1, CE02, CE03, CE04, CE05, CSO1, CS02, CS06, RE01, RE02, RE02.1, RSO1, and RS03.1.

3.4.8 Interchangeability (first article inspection only). All subsystem parts having identical part number shall be functionally and dimensionally interchangeable or replaceable.

3.5 UTS acceptance test. Each UTS shall be functionally tested in accordance with 4.5.1 to assure compliance with specified operating acceptance requirements defined in 3.4.1.

3.6 Proof test (first article inspection only). The UTS shall operate in accordance with the intent of the design when assembled to an accepted AH-IS series helicopter.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article by the contractor shall be as specified in the contract.

3.8 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner; and all manufacturing, processing, and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, cracks, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous, however, a few light scratches not exposing base material may be permitted. All required markings and stamping shall be neat and sharply defined. Required packing shall be dry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-W-63150 shall apply and are hereby made a part of this detail specification.

4.1.1 Certification of provisions. The certification provisions of MIL-W-63150 shall apply (see 6.2.1).

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3).

b. Quality Conformance Inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Pressure Transducer	C116C3555	2
Gun Control Assembly	F11830418	2
Slider Assembly	C11830565	2
Adapter, Recoil Mechanism	D11830567	2
Feed Assembly, Ammo	F11830886	2
Turret Control Assembly	F11830908	2
Logic Control Unit Assembly	F11830922	2
Turret Assy	F11830962	2
Drive Assembly	F189F734	2
Drive Motor Assembly	D905D490	2
Universal Turret Subsystem	F11830963	2

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4.3.2 Inspections to be performed. See Table III specified herein.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen in the sample to comply with any of the stated requirements.

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TABLE III. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING
	Universal Turret Subsystem				See
					NEXT HI
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRA
	Pressure Transducer (Dwg. C116C3555) Examination for defects	2		3.2	4.4.
	Gun Control Assembly (Dwg. F11830418) Examination for defects	2		3.2	4.4.
	Slider Assembly (Dwg. C11830565) Examination for defects	2		3.2	4.4.
	Adapter, Recoil Mechanism (Dwg. D11830567) Examination for defects	2		3.2	4.4.
	Feed Assembly, Ammo (Dwg. F11830886) Examination for defects	2		3.2	4.4.
	Turret Control Assembly (Dwg. F11830908) Examination for defects	2		3.2	4.4.

NOTES:

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TABLE III. First article inspection.

MIL

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWIN
Cont'd	Universal Turret Subsystem				See NEXT H
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGR
	Logic Control Unit Assembly (Dwg. F11830922) Examination for defects	2		3.2	4.4
	Turret Assembly (Dwg. F11830962) Examination for defects	2		3.2	4.4
	Drive Assembly (Dwg. F189F734) Examination for defects	2		3.2	4.4
	Motor, Elec, Drive (Dwg. D905D490) Examination for defects	2		3.2	4.4
	Universal Turret Subsystem (Dwg. F11830963) Examination for defects	2		3.2	4.4
	Environmental Test	(a)		3.4.5	4.5
	Interchangeability	(a)		3.4.8	4.6
	Proof test	(a)		3.6	4.5

NOTES: (a) This test shall be performed on the same samples subjected to th for defects.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with the requirements. In addition, each lot shall contain:

a. Gun Control Assemblies of one lot interfix number from one manufacturer and complying with all requirements in MIL-T-63520(AR).

b. Slider Assemblies of one lot interfix number from one rer and complying with all requirements in MIL-S-63249(AR).

c. Recoil Mechanism Adapters of one lot interfix number from one manufacturer and complying with all requirements in MIL-A-63425(AR).

d. Ammo Feed Assemblies of one lot interfix number from one manufacturer and complying with all requirements in MIL-F-63428(AR).

e. Turret Control Units of one lot interfix number from one manufacturer and complying with all requirements in MIL-T-63534(AR).

f. Logic Control Assemblies of one lot interfix number from one manufacturer and complying with all requirements in MIL-T-63536(AR)

g. Turret Assemblies of one lot interfix number from one manufacturer.

h. Gun Drive Assemblies of one lot interfix number from one manufacturer and complying with all requirements in MIL-D-63516(AR).

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i. Electric Drive Motors of one lot interfix number from one manufacturer and complying with all requirements in MIL-T-63535(AR).

4.4.2 Examination. The examinations listed in this specification for Classification of Defects shall be performed on inspection lots as defined in 4.4.1 of this specification.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER C116C3555 (2 shts)
4.4.2.1	Transducer, Pressure				NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Performance requirements		--	3.2	4.1.1
102	Environmental requirements		--	3.2	4.1.1
103	PITOT tube pressure below static pressure *(Note 11)		0.65%	3.2	4.7
104	Diameter under flange, max.		0.65%	3.2	Gage
105	Total height, max (4.10)		0.65%	3.2	Gage
106	Part damaged (cracked, dented, etc)		0.65%	3.2	Visual
107	Orifice missing *(Note 9)		0.65%	3.2	Visual
108	Assembly improper		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Poor workmanship		2.50%	3.8	visual

NOTES:

* Notes refer to drawing.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER F11830418 (2 shts)
4.4.2.2	Gun Control Assembly				NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering improper		--	3.2	4.1.1
102	Part missing		0.65%	3.2	Visual
103	Part damaged (cracked, dented, etc)		0.65%	3.2	Visual
104	Assembly improper		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Poor workmanship		2.50%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63533 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Slider Assembly				C11830565
					NEXT HIGHER ASSEMBLY
					F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Grease missing from housing cavity		0.65%	3.2	Visual
102	Parts missing		0.65%	3.2	Visual
103	Parts damaged (cracked, dented, etc)		0.65%	3.2	Visual
104	Assembly improper (improper orientation)		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Poor workmanship		2.50%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Adapter, Recoil Mechanism				C11830567
					NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Grease missing		0.65%	3.2	Visual
102	Part missing		0.65%	3.2	Visual
103	Part damaged (cracked, dented, etc)		0.65%	3.2	Visual
104	Assembly improper (improper orientation)		0.65%	3.2	Visual
105	Safety wire improperly assembled or missing		0.65%	3.2	Visual
Minor					
201	Torque of mounting screws improper		1.50%	3.2	SMTE
202	Marking misleading, illegible or missing		1.50%	3.2	Visual
203	Poor workmanship		2.50%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63533 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER F11830886
4.4.2.5	Feed Assy, Ammo				NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Part missing		0.65%	3.2	Visual
102	Part damaged (cracked, dented, etc)		0.65%	3.2	Visual
103	Assembly improper		0.65%	3.2	Visual
104	Assembly cables improperly connected		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Poor workmanship		2.50%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER F11830908
4.4.2.6	Turret Control Assy				NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering improper		--	3.2	4.1.1
102	Assembly improper		0.65%	3.2	Visual
103	Part missing		0.65%	3.2	Visual
104	Part damaged (cracked, dented, etc)		0.65%	3.2	Visual
Minor					
201	Torque improper *(Note 4)		1.50%	3.2	SMTE
202	Marking misleading, illegible or missing		1.50%	3.2	Visual
203	Poor workmanship		2.50%	3.8	Visual

NOTES:

* Notes refer to drawing

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63533 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Logic Control Unit Assy				F11830922
					NEXT HIGHER ASSEMBLY
					F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering improper		--	3.2	4.1.1
102	Assembly improper		0.65%	3.2	Visual
103	Part missing		0.65%	3.2	Visual
104	Part damaged (cracked, dented, etc)		0.65%	3.2	Visual
Minor					
201	Torque of screws improper		1.50%	3.2	SMTE
202	Silicone compound missing from required areas		1.50%	3.2	Visual
203	Marking misleading, illegible or missing		1.50%	3.2	Visual
204	Identification plate missing		1.50%	3.2	Visual
205	Poor workmanship		2.50%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-T-63533(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER F11830962 (6 shts)
4.4.2.8	Turret Assembly				NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Adjusting of brackets improper *(Notes 3 and 4)		0.65%	3.2	4.8
102	Adjusting switch improper *(Note 8, 10 and 15)		0.65%	3.2	4.9
103	Shimming improper *(Note 2, 7, 15, 20 and 23)		0.65%	3.2	SMTE
104	Degrees elevation improper (Note 15)		0.65%	3.2	SMTE
105	Degrees azimuth improper (Note 15)		0.65%	3.2	SMTE
106	Link preload adjustment improper *(Note 15)		0.65%	3.2	SMTE
107	Press fitting of dowel pins improper *(Note 21)		0.65%	3.2	SMTE
108	Installing proximity switch improper		0.65%	3.2	SMTE
109	Cable installation improper *(Notes 31 throughout 41)		100%	3.2	Visual
110	Installation improper *(Note 24)		100%	3.2	Visual
111	Grease missing from gear teeth bearings and areas indicated on dwg. *(Note 1)		0.65%	3.2	Visual

NOTES:

* Notes refer to drawing

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.8	Turret Assembly				F11830962 (2 shts)
					NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Holes (2) in anti-backlash gear not in line when meshed with mating gear *(Note 5)		0.65%	3.2	Visual
113	Touch up improper		0.65%	3.2	Visual
114	Wrapping of excess cable improper *(Note 9)		0.65%	3.2	Visual
115	Staking of dowel pins improper *(Note 11)		0.65%	3.2	Visual
116	Marking misleading, illegible, or missing		0.65%	3.2	Visual
117	Final orientation of cable clamps improper *(Note 13)		0.65%	3.2	Visual
118	Adhesive to bond shims to upper support improper *(Note 16)		0.65%	3.2	Visual
119	Strapping of harness assembly improper *(Note 17)		0.65%	3.2	Visual
120	Safety wiring improper or missing *(Notes 18, 19 and 22)		0.65%	3.2	Visual
121	Surface receiving item improper *(Note 25)		0.65%	3.2	Visual

NOTES:
* Notes refer to drawing

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MI L-T-63533 (AR)

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.8	Turret Assembly		F11830962 (2 shts)
			NEXT HIGHER ASSEMBLY F11830963

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
122	Identification plate missing *(Note 26)		0.65%	3.2	Visual
123	Part missing		0.65%	3.2	Visual
124	Part damaged (cracked, dented, etc)		0.65%	3.2	Visual
125	Assembly improper		0.65%	3.2	Visual
Minor					
201	Poor workmanship		2.50%	3.8	Visual

NOTES:

* Notes refer to drawing

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QUALITY CONFORMANCE INSPECTION

CIASSIFICATJON OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER F189F734
4.4.2.9	Drive Assy, Gun				NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Grease missing from gear teeth		0.65%	3.2	Visual
102	Parts missing		0.65%	3.2	Visual
103	Parts damaged		0.65%	3.2	Visual
104	Assembly improper		0.65%	3.2	Visual
Minor					
201	Nut adjusted improperly *(Note 3)		1.50%	3.2	SMTE
202	Torque of fasteners improper		1.50%	3.2	SMTE
203	Marking misleading, illegible or missing		1.50%	3.2	Visual
204	Touch-up missing from unprotected surfaces		1.50%	3.2	Visual
205	Poor workmanship		2.50%	3.8	Visual

NOTES: * Notes refer to drawing

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63533(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER D905D490 (2 shts)
4.4.2.10	Motor, Elec, Drive				NEXT HIGHER ASSEMBLY F11830963
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Performance requirements *(Notes 1, 11, and 12)		--	3.2	4.1.1
102	Mechanical characteristics *(Note 9)		--	3.2	4.1.1
103	Notched tangs not removed after installation *(Note 7)		0.65%	3.2	Visual
Minor					
201	Marking misleading, illegible or missing		1.50%	3.2	Visual
202	Poor workmanship		2.50%	3.8	Visual

NOTES:

* Notes refer to drawing

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-T-63533 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11830963
4.4.2.11	Universal Turret Subsystem				NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Acceptance test		100%	3.5	4.5.1
102	Parts missing		0.65%	3.2	Visual
103	Parts damaged (cracked, dented, etc)		0.65%	3.2	Visual
104	Assembly improper		0.65%	3.2	Visual
105	Poor workmanship		0.65%	3.8	Visual
Minor					
201	None defined				

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). All inspection and test equipment shall be calibrated in accordance with MIL-W-63150. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 herein.

4.5 Methods of inspection.

4.5.1 Acceptance test plan. An acceptance test plan and its inspection design equipment shall be submitted by the contractor to the procuring agency for approval.

4.5.2 Environmental characteristics (first article inspection only).

4.5.2.1 Vibration test.

a. Gun turret. The gun turret, including all turret mounted electrical/electronic components, and each of the weapons, shall be installed in a test fixture in a manner that will simulate service usage and subjected to a vibration test in accordance with MIL-STD-810, Method 514.2, Procedure I, Part 1, Figure 514.2-3, (Curve M) and Table 514.2-III, Vibration levels may be reduced per Note 1 of Table 514.2-III.

b. Electronic control units. The electronic control units shall be installed in a test fixture in a manner that will simulate service usage and subjected to a vibration test in accordance with MIL-STD-810, Method 514.2, procedure I, Part 1, Table 514.2-111, per the curve presented in Figure 4. The time for cycling from 5 to 500 and back to 5 Hz shall be not less than 15 minutes.

c. Resonance search and dwell. Resonance dwell for items above shall be confined to resonances occurring within the frequency ranges shown below, using paragraph 4.5.1.1 and 4.5.1.2 of M1L-STD-810, Procedure 514.2. Resonance dwells shall be for 30 minutes each and can be counted toward cycling time. If resonances

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Resonances Occurring Between	Resonant Dwell Frequency
4 - 100 Hz	At resonance

d. Transportation vibration testing shall be in accordance with MIL-STD-810, Method 514.2, Procedure X.

4.5.2.2 Shock test.

a. Gun turret. The gun turret, including all turret-mounted electrical/electronic components, and each of the designated weapons, shall be installed in a test fixture that will simulate service usage and subjected to a shock test in accordance with MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2, for flight vehicle equipment. The “g” value shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter.

b. Electronic control units. The electronic control units shall be installed in a test fixture that will simulate service usage and subjected to shock tests in accordance with:

(1) MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-2 for flight vehicle equipment. The “g” value shall be within + 10 percent when measured with a 0.2 to 250 Hz filter.

(2) MIL-STD-810, Method 516.2 Procedure V.

4.5.2.3 Temperature-altitude-humidity.The UTS shall be subjected to temperature-altitude-humidity testing as follows:

a. With the UTS nonoperating, adjust test chamber temperature to -65°F and stabilize test specimen. Lower chamber internal pressure to an equivalent of 50,000 feet altitude and maintain temperature and altitude for not less than 3 hours.

b. With the UTS nonoperating and temperature at -65°F, adjust internal chamber pressure to 20,000, stabilize chamber and operate UTS.

c. Readjust test chamber internal pressure to room ambient and allow chamber internal temperature to rise to 0°F.

d. Allow frost and some icing to form on UTS and when UTS is completely covered with frost and some icing, operate UTS.

e. Allow UTS to return to room ambient temperature and stabilize and operate UTS.

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4.5.2.4 Temperature-humidity. The UTS shall be subjected to temperature-humidity testing as follows:

a. With the UTS nonoperating, adjust chamber temperature at +160°F with uncontrolled humidity and stabilize UTS. Maintain this condition for not less than 4 hours.

b. Adjust chamber internal relative humidity to 95 ± 5 percent relative humidity. While maintaining chamber internal temperature at +125°F, stabilize chamber and test specimen.

c. Operate UTS while maintaining +125°F temperature and 95 ± 5 percent relative humidity.

d. Return chamber to room ambient (temperature and humidity). Upon stabilization of UTS, operate UTS.

4.5.2.5 Salt fog. The equipment mounted external to the aircraft, including gun-control box, shall be subjected to the test of Method 509.1 of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

4.5.2.6 Sand dust. The equipment shall be subjected to the test of Method 510.1 of MIL-STD-810, Procedure 1, except temperature to be +73°F and relative humidity of less than 22 percent. Air velocity to be 1750 ± 250 feet per minute and dust concentration at 0.3 +0.2 grams per cubic feet (gm/ft^3) and maintain these conditions for 6 hours. After 6 hours, turn off air velocity and dust feed and stabilize. Perform operation test before any cleaning is performed.

4.5.2.7 Fungus. The equipment shall be subjected to the test of Method 508.1 of MIL-STD-810, Procedure I. Gasket sealed or other nonhermetic sealed test items shall be opened during test exposures and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required, but examination of the concentration after exposure period shall reveal no evidence of visible fungus on any surfaces.

4.5.2.8 Explosive atmosphere. The equipment shall be subjected to the test of Method 511.1 of MIL-STD-81O, Procedure I, except test to be limited to maximum altitude of 20,000 feet.

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4.5.2.9 Rain. Method 506.1, Procedure I, of M1L-STD-810 shall apply. During each 30 minute rain period, the wind shall be maintained for not less than 15 minutes commencing 5 minutes after the start of the period as specified in the procedures. The equipment shall be operated during the last 30 minutes of the 30-minute period.

4.5.2.10 EMC tests. The subsystem shall be installed in laboratory test fixtures in a manner that will simulate actual service usage and tested to MIL-STD-461, Methods CEO1, CE02, CE03, CE04, CE05, CSO1, CS02, CS06, RE02, RE02.1, RSO1 and RS03.1.

4.5.3 Proof test (first article inspection only). When specified in the contract (see 6.2) two (2) UTS'S shall be forwarded to the site specified by the contracting agency and assembled to an accepted modernized COBRA Helicopter AH-IS. The UTS shall be tested for compliance with 4.5.1 and the applicable specification requirements of boresighting and reliability. The UTS shall function in accordance to the intent of the design and with no evidence of malfunction and satisfactory to the procuring agency. All test results shall be forwarded in accordance with 6.4.

4.6 Interchangeability testing.

4.6.1 Universal Turret Subassembly (UTS). A sample of 2 UTS'S, randomly selected, shall be tested for interchangeability using the test method specified in 4.6.3. UTS's taken for interchangeability testing shall have been found satisfactory in all other examinations and tests. The UTS's shall comply with the applicable specification functioning tests of 4.5.1, before and after the interchange of parts. It shall be the responsibility of the agency responsible for the test to inform the contracting officer by letter or other methods satisfactory to the contracting officer, when selective assembly has become necessary. Statement shall contain as a minimum, the-contractor's name and FSCM, contract number, parts involved in the selective fit, conditions of parts and rationale for action taken. No hand refinement of parts shall be permissible without approval by the contracting officer. No malfunctions shall be allowed in the functioning tests. Failure of UTS in the interchangeability test shall cause retest or rejection of the first article. At the discretion of the Government repre-sentative, an interchangeability retest may be allowed. Failure during the retest shall cause rejection of the first article. A sample of 4 UTS'S from each retest shall be tested using the same procedure described below.

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4.6.2 Repair parts. At least 2 parts of concurrent repair parts shall be subjected to the interchangeability test specified in 4.6.3. Failure of any part to meet the interchangeability requirements shall be cause for rejection, subject to reconditioning and further test. A sample consisting of double the number of parts used in the original test shall be tested using the test methods specified above.

4.6.3 Test method for interchangeability. UTS's shall be tested by disassembling and then reassembling the parts specified below. Parts shall be disassembled from the UTS'S and identical parts placed together and mixed. Any commercial part (screws, pins, etc) rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test.

LIST OF INTERCHANGEABLE PARTS

TITLE	DRAWING
Pressure Transducer	C116C3555
Gun Control Assembly	F11830418
Slider Assembly	C11830565
Adapter, Recoil Mechanism	D11830567
Feed Assembly, Ammo	F11830886
Turret Control Assembly	F11830908
Logic Control Unit Assembly	F11830922
Turret Assembly	F11830962
Drive Assembly	F189F734
Drive Motor Assembly	D905D490

4.6.4 Repair parts. Concurrent repair parts shall be tested by disassembling a UTS previously tested 4.6.3 and then reassembling them using the concurrent repair parts. No hand refinement of parts will be allowed and the assembled UTS shall operate and function as intended. This test may be performed independently of the UTS interchangeability testing specified in 4.6.1 and at more frequent intervals using accepted UTS'S taken from current production.

4.7 PITOT tube pressure. PITOT tube pressure must be at the same level or above static pressure when tested. Test equipment and procedures shall be satisfactory to the contracting officer. (See 4.4.4).

4.8 Turret assembly brackets. Turret assembly brackets should be adjusted as specified on the applicable drawing when actuating the azimuth or elevation switches. Proper clearence between actuator and switch, and actuator and support shall be inspected. Test equipment and procedures shall be satisfactory to the contracting officer. (See 4.4.4).

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4.9 Turret assembly switches. Turret assembly switches shall be adjusted as specified on the applicable drawing. Test equipment and procedures shall be satisfactory to the contracting officer. (See 4.4.4).

5. PACKAGING

5.1 Packaging. The components or assemblies of the subsystem shall be packaged for shipping in accordance with PDS 11830963.

6. NOTES

6.1 Intended use. The UTS covered by this specification is intended for use with Gun, automatic, 20MM: M197 on Armament Subsystem, Helicopter, 20MM: M97E1 and M97E2.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 procurement requirements.

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

e. First Article sample requirements, if other than specified in Section 4 of this specification.

f. Packaging requirements, if other than specified in Section 5 of this specification.

g. Certificates of conformance for each lot or shipment of product.

6.2.2 contract data requirements. Contract data requirements for inspection equipment design shall conform to Data Item Description DI-R-1714, Tailored.

6.3 Submission of inspection equipment design for approval. See MIL-W-63150. Equipment designs shall be submitted as required to: Commander, ARRADCOM, ATTN: DRDAR-QAF-1, Dover, NJ 07801. Request letter of submittal shall state contractor, contract number, specification number, item nomenclature, and classification of defects and test paragraphs.

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6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commander, ARRADCOM, ATTN: DRDAR-QAF-S, Dover, NJ 07801.

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☐ VENDOR ☐ USER ☐ MANUFACTURER

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OF VALIDATION

MIL-T-63533(AR)
NOTICE 1
22 MARCH 1988

MILITARY SPEICIFICATION

TURRET SUBSYSTEM, UNIVERSAL (UTS) FOR ARMAMENT SUBSYSTEM, HELICOPTER, 20mm: M97El AND M97E2

MIL-T-63533 (AR), dated 27 December 1981, has been reviewed and determined to be Valid for use in acquisition.

Preparing activity:
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